LAB VI: EVALUATE DIFFERENT ARITHMETIC OPERATIONS AND LOGICAL OPERATIONS ON TWO 32-BIT DATA USING ARM PROCESSOR

1. AIM OF THE EXPERIMENT

EVALUATE DIFFERENT ARITHMETIC OPERATIONS AND LOGICAL OPERATIONS ON TWO 32-BIT DATA USING ARM PROCESSOR

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2. OBJECTIVES

1. Perform Addition and Subtraction of two 32-bit numbers using data processing addressing mode (with immediate data).
2. Perform Addition, Subtraction, and Multiplication of two 32-bit numbers using load/store addressing mode.
3. Perform the logical operations (AND, OR, XOR, and NOT) on two 32-bit numbers using load/store addressing mode.

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3. THEORY

ARM Processor:

ARM (Advanced RISC Machines) is a 32-bit RISC processor with Load/Store architecture where instructions operate on registers only. Key features: 3-operand instructions, large register set, pipelined execution, and conditional execution capability.

Data Sizes: Byte (8 bits), Halfword (16 bits), Word (32 bits)

Registers:

- R0-R12: General purpose registers
- R13 (SP): Stack Pointer, R14 (LR): Link Register, R15 (PC): Program Counter
- CPSR: Current Program Status Register with flags N, Z, C, V
- SPSR: Saved Program Status Register

Condition Flags (CPSR):

N (Negative), Z (Zero), C (Carry), V (Overflow) - updated by arithmetic/logical operations

Addressing Modes:

1. **Data Processing:** Register (ADD R0, R1, R2), Immediate (ADD R3, R3, #1), Scaled Register (ADD R0, R0, R0, LSL #2)

2. **Load/Store:** Immediate offset (LDR R0, [R1, #4]), Register offset (LDR R0, [R1, R2]), Scaled register (LDR R0, [R1, R2, LSL #2])
3. **Branch:** 24-bit immediate value for target address
4. **Load/Store Multiple:** IA (Increment After), IB (Increment Before), DA (Decrement After), DB (Decrement Before)

Instruction Categories:

1. **Data Processing:** Arithmetic (ADD, SUB, MUL), Logical (AND, ORR, EOR, BIC), Move (MOV, MVN), Compare (CMP, CMN, TST, TEQ), Shift/Rotate (LSL, LSR, ASR, ROR)
2. **Data Transfer:** LDR/STR (word), LDRB/STRB (byte), LDRH/STRH (halfword), LDM/STM (multiple)
3. **Control Flow:** B (unconditional), BEQ/BNE (equal/not equal), BLT/BGT/BLE/BGE (comparisons), BL (branch with link)

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4. OBSERVATION

Objective 1: Perform Addition and Subtraction of two 32-bit numbers using data processing addressing mode (with immediate data).

a) Pseudocode:

START

LOAD first 32-bit immediate value into register R0

LOAD second 32-bit immediate value into register R1

ADDITION:

ADD contents of R0 and R1

STORE the result in register R2

SUBTRACTION:

SUBTRACT contents of R1 from R0

STORE the result in register R3

DISPLAY results stored in registers R2 and R3

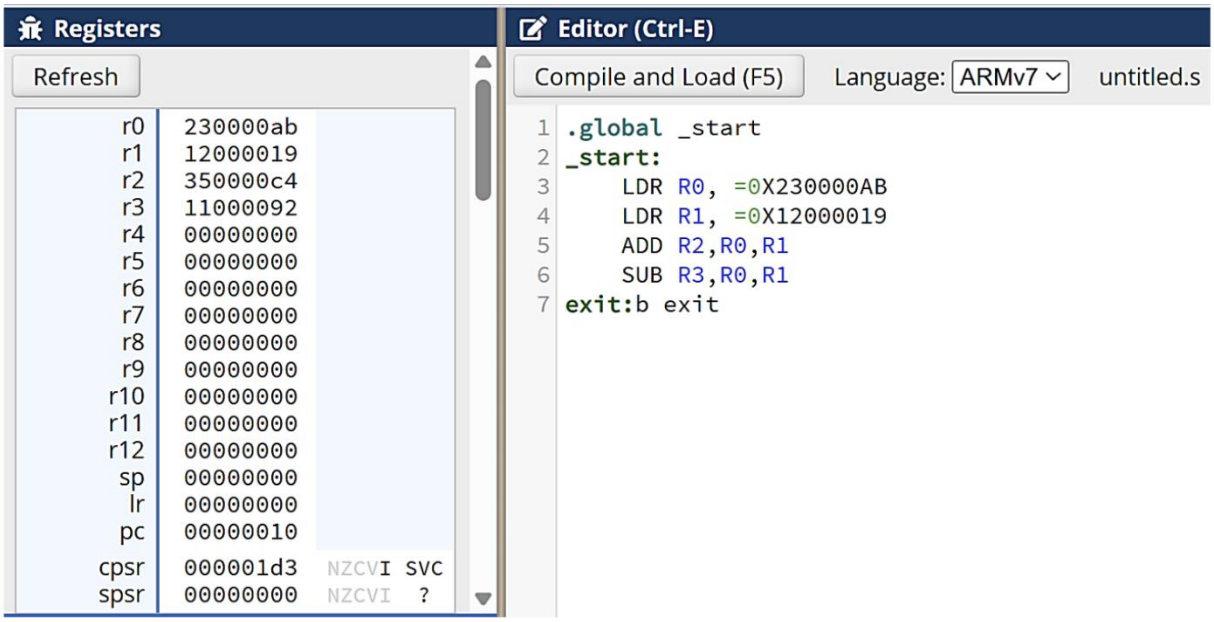
STOP

END

b) Code:

```
.global _start
_start:
    LDR R0, =0X230000AB
    LDR R1, =0X12000019
    ADD R2,R0,R1
    SUB R3,R0,R1
exit:b exit
```

c) Output:



d) Analysis:

Input		Output	
Memory Location	**Data**	**Memory Location**	**Data**
-	0X230000AB	-	0X350000C4
-	0X12000019	-	0X11000092

Objective 2: Perform Addition, Subtraction, and Multiplication of two 32-bit numbers using load/store addressing mode.

a) Pseudocode:

START

INITIALIZE a memory pointer to the base address

LOAD first 32-bit number from memory into register R1

INCREMENT memory pointer to next memory location

LOAD second 32-bit number from memory into register R2

INCREMENT memory pointer to next memory location

ADDITION:

ADD contents of R1 and R2

STORE the result into memory

INCREMENT memory pointer

SUBTRACTION:

SUBTRACT contents of R2 from R1

STORE the result into memory

INCREMENT memory pointer

MULTIPLICATION:

MULTIPLY contents of R1 and R2

STORE the result into memory

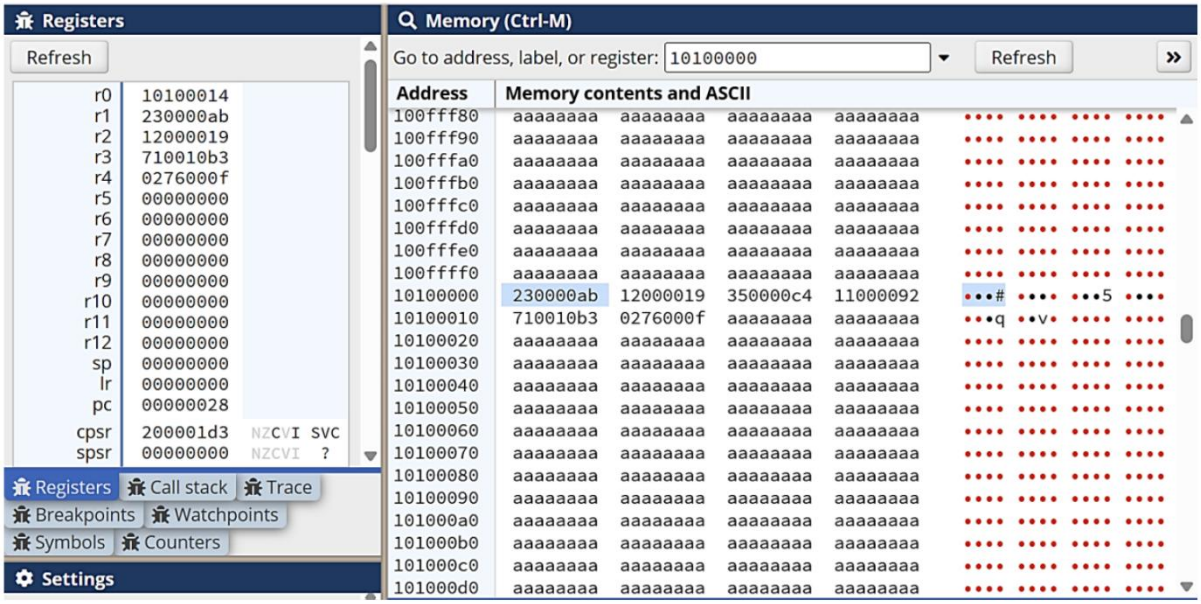
STOP

END

b) Code:

```
.global _start
_start:
    LDR R0, =0X10100000
    LDR R1,[R0],#4
    LDR R2,[R0],#4
    ADD R3,R1,R2
    STR R3,[R0],#4
    SUB R4,R1,R2
    STR R4,[R0],#4
    MUL R5,R1,R2
    STR R5,[R0]
exit:b exit
```

c) Output:



d) Analysis:

Input		Output	
Memory Location	**Data**	**Memory Location**	**Data**
0X10100000	0X230000AB	0X10100008	0X350000C4
0X10100004	0X12000019	0X1010000C	0X11000092
-	-	0X10100010	0X710010B3
-	-	0X10100014	0X0276000F

Objective 3: Perform the logical operations (AND, OR, XOR, and NOT) on two 32-bit numbers using load/store addressing mode. a) Pseudocode:

START

INITIALIZE memory pointer to base address

LOAD first 32-bit number from memory into register R1

INCREMENT memory pointer to next memory location

LOAD second 32-bit number from memory into register R2

INCREMENT memory pointer to next memory location

AND OPERATION:

PERFORM bitwise AND on R2 and R1

STORE the result into memory

INCREMENT memory pointer

OR OPERATION:

PERFORM bitwise OR on R2 and R1

STORE the result into memory

INCREMENT memory pointer

XOR OPERATION:

PERFORM bitwise XOR on R2 and R1

STORE the result into memory

INCREMENT memory pointer

NOT OPERATION:

PERFORM bitwise NOT on R1

STORE the result into memory

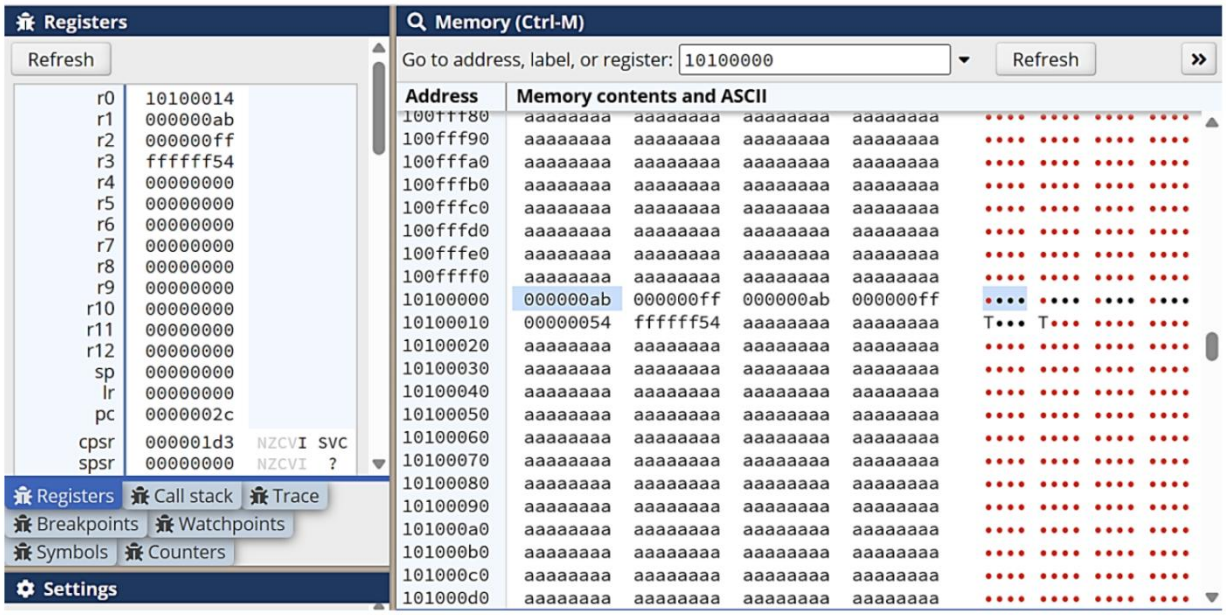
STOP

END

b) Code:

```
.global _start
_start:
    LDR R0, =0X10100000
    LDR R1,[R0],#4
    LDR R2,[R0],#4
    AND R3,R2,R1
    STR R3,[R0],#4
    ORR R4,R2,R1
    STR R4,[R0],#4
    EOR R5,R2,R1
    STR R5,[R0], #4
    MVN R6, R1
    STR R6, [R0]
exit:b exit
```

c) Output:



d) Analysis:

Input		Output	
Memory Location	**Data**	**Memory Location**	**Data**
0X10100000	0X000000AB	0X10100008	0X000000AB
0X10100004	0X000000FF	0X1010000C	0X000000FF
-	-	0X10100010	0X00000054
-	-	0X10100014	0XFFFFFF54

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5. CONCLUSION

Post Lab:

1. Give any examples of five arithmetic and logical instructions.

2. Differentiate between LDR and STR instruction.

3. Which of the following instructions is not valid a) MOV R7.R2 b) LDR R1, =LABEL

4. Explain briefly condition codes (flags) of ARM processor.

5. Which condition codes (flags) is considered for the following branch instructions?

B Label

BEQ label

BLT label

6. If the registers r1,r2,r3 contain the values 10,20,30 respectively, what will be the value in register r4 after the execution of the following code segment?

ADD r4, r1, r3

SUBS r4, r2, r4

RSB r4, r1, r4